How aggregation can conceal composition: aggregate biocapacity and the identifiability of modeled ecological-capital drawdown

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Abstract

Total biocapacity aggregates heterogeneous land accounts whose yields and regeneration rates differ. We ask what this headline reveals about land composition. We formulate a two-land delay-differential model in which account shortfall triggers one-to-one conversion from ecological-capital to fast-provisioning area; capital quality evolves separately, and degradation acts through accumulated ecological debt. Three propositions are proved and checked numerically. (i) During active conversion, the conversion contribution to biocapacity is positive exactly when the fast-land yield exceeds the per-hectare value of capital land; area A_c declines, while an aggregate-only observation cannot identify this composition. (ii) Under frozen demand, the balance point is unique. Above the demand ceiling $E_{\mathrm{ceil}} \approx 1.14\,\mathrm{gha}$, capital-land area approaches its typed floor asymptotically. Affine deficit-side dynamics give a constant leading eigenvalue ($-0.054\,\mathrm{yr}^{-1}$), so the boundary collision has no fold, hysteresis, or critical slowing down. With endogenous demand, drawdown depth varies continuously with initial demand. (iii) Technology gain is parameter-bounded, whereas debt erosion is bounded by sustained shortfall. Applied to the National Footprint and Biocapacity Accounts (1961–2022), an exact Divisia decomposition finds aggregate biocapacity up 23%, cropland biocapacity up 2.85-fold, non-cropland biocapacity nearly flat, and per-capita biocapacity halved. NFA categories do not directly measure modeled area A_c or quality q. These model results are conditional, not estimates of a global physical-stock trend; we propose a physical-stock scoreboard with explicit units and valuation.

Keywords: biocapacity, ecological footprint, land-use conversion, aggregation, identifiability, critical slowing down, delay differential equations, carrying capacity

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1. Introduction

Carrying capacity is one of ecology's most enduring and most elusive ideas. In classical population biology it is a fixed ceiling; imported into global human ecology it is usually treated as a static boundary, the maximum number of people the Earth can support (cf. Cohen, 1995). Treating the ceiling as fixed overlooks a feedback that matters: a population that overshoots its resource base degrades the base and lowers the future ceiling. This paper is concerned with a subtler failure than a fixed ceiling. It concerns the *measurement* on which monitoring and policy depend.

The standard aggregate measure of annual regenerative supply is biocapacity, expressed in the National Footprint and Biocapacity Accounts (NFA) in global hectares (gha) for each annual account. Under the Global Footprint Network (GFN) convention, annual Ecological Footprint and biocapacity values are reported in gha, not gha yr^{-1} (Global Footprint Network, 2009, 2019). A global hectare is a productivity-normalized hectare whose value incorporates yield and equivalence factors (Borucke et al., 2013; Lin et al., 2018). Because the weighting sits *inside* the unit, an aggregate biocapacity series does not tell a monitor whether capacity rose because land became more productive, because more land was brought into provision, or because ecological capital was converted into provisioning land. The composite-indicator literature warns that additive aggregation turns weights into trade-offs, so that a deficit in one dimension can be offset by a surplus in another (Nardo et al., 2008; Becker et al., 2017; Fischer et al., 2022); biocapacity is exactly such a composite, and the footprint framework has been criticized on related grounds (Blomqvist et al., 2013; Giampietro and Saltelli, 2014; van den Bergh and Grazi, 2015; Galli et al., 2016). The aggregate remains useful as a headline, but only if its components are reported alongside it. What has been missing is a dynamical account of *how* a rising aggregate and a falling capital base can coexist, for how long, and what would be needed to tell them apart.

Coupled human–environment models have largely treated the environment as a single stock or a single aggregate balance: the World3 system (Meadows et al., 1972, 1974, 2004), the Ricardo–Malthus models of Easter Island (Brander and Taylor, 1998; Anderies, 2000), and more recent bidirectionally coupled Earth-system frameworks (Motesharrei et al., 2016; Donges et al., 2017; Calvin and Bond-Lamberty, 2018; Henderson and Loreau, 2018). Estimates of Earth's carrying capacity span several orders of magnitude (Co-

hen, 1995; van den Bergh and Rietveld, 2004), some authors reject the notion for humans altogether (Simon, 1996), and others let the ceiling itself vary logistically (Meyer, 1994; Meyer and Ausubel, 1999) or treat natural capital as a depreciating asset (Dasgupta, 2021). Demographic projections used in most global assessments assume that population unfolds independently of environmental constraints (United Nations, 2022). Within this literature the environment has one book; the question of what an aggregate over two books with different yields and regeneration times can conceal has not been posed as a modeling problem, although the land-use literature has documented the historical operator – conversion of forest, grassland and wetland into cropland – in detail (Haberl et al., 2007; Krausmann et al., 2013; Erb et al., 2017).

Consider an orchard with two kinds of ground. Fast provisioning land – cropland and intensively managed systems – yields much and is quickly restored, but has little standing regenerative value. Ecological-capital land – forest, soil, wetland, long-rotation systems – has a large standing value and a regenerative flow that is slowly rebuilt. Demand is evaluated against the human-available annual account total; any remaining account shortfall triggers conversion from capital land into fast provision. Because aggregate biocapacity is a value-weighted sum over the two land books, conversion can raise the account total while capital-land area A_c shrinks. A harvest that looks stable, or a rising aggregate measure of capacity, can therefore mask a shrinking ecological-capital account. Neither the orchard nor its harvesters respond instantly – forest loss, soil degradation and demographic adjustment unfold over decades – which is why societies can prosper for long periods while quietly undermining their productive base, only to decline abruptly (Tainter, 1988; Diamond, 2005). The companion ledger work calls the general phenomenon the *productivity illusion* (Abaee, 2026c); the specific form addressed here is the *composition* illusion, in which the aggregate rises not because technology outpaces degradation but because the books are reweighted.

We address three questions. **(Q1)** Under what conditions does conversion of ecological-capital land raise the aggregate while capital-land area A_c falls, how long does the mask last, and can the composition be identified from the aggregate? **(Q2)** Is the resulting capital drawdown a catastrophic shift in the sense of the alternative-stable-states literature (Scheffer et al., 2001), with a fold, hysteresis and early-warning signals, or something structurally different? **(Q3)** Is the mask self-limiting, and what does recovery require?

The paper makes four contributions. First, it formulates a minimal two-

land model (Section 2) in which area is conserved one-to-one across three land books, per-hectare capital quality is a separate state, conversion is a rate with an explicit time constant, and degradation acts on the surviving yield through an accumulated ecological debt; the model is a delay differential equation with ecological (regeneration) and demographic delays (Kuang, 1993; Hutchinson, 1948). Second, it proves three propositions (Section 3): a composition identity that gives the exact sign condition for the conversion contribution and shows that an aggregate-only observation does not locally identify the capital loss (Proposition 1); the uniqueness of the frozen-demand balance point, the location of the demand ceiling, and the absence of a fold and of critical slowing down, derived from the affine structure of the deficit-side dynamics (Proposition 2); and a boundedness asymmetry between the technology and debt channels that makes the mask transient (Proposition 3). Third, it applies the composition identity to the measured NFA series for 1961–2022 with an exact logarithmic-mean (Divisia) decomposition and the measured land-type accounts (Section 3.5). Fourth, it states the falsifiable content of the model as five predictions and draws the reporting implication for sustainability indicators (Sections 3.6 and 4). The model is deliberately simple and most of its parameters are representative rather than fitted; it is not a forecast. Its purpose is to make the composition logic of aggregate biocapacity explicit and to state which outcomes are structural and which are conditional on effective parameters that are not separately identified.

2. Methods

2.1. A two-land model with conserved area and a separate quality state

The bioproductive area A_{tot} is divided into three books: fast-provisioning land A_f (cropland and intensively managed systems), ecological-capital land A_c (forest, soil, wetland and long-rotation systems, consolidated into one effective class), and a reserve A_r (fallow, retired or not yet allocated land), with $A_f + A_c + A_r = A_{tot}$ at all times. Capital land carries a per-hectare quality index $q \in [0, q_{\max}]$ (a capacity state, not an area), which regenerates logistically,

$$G_c(q) = \rho_c \, q \left(1 - \frac{q}{q_{\max}}\right), \tag{1}$$

so that the per-hectare value of capital land, in annual NFA-account-yield units, is

$$\Upsilon_c(q) = b_c + b_{G,c} \, G_c(q), \tag{2}$$

the sum of a direct per-hectare yield b_c and the regeneration-value coefficient $b_{G,c}$ times the quality growth $G_c(q)$. Biocapacity is the additive two-book aggregate

$$B = Y_f + Y_c = b_f A_f + \Upsilon_c(q)\, A_c, \tag{3}$$

in gha, the annual NFA account unit (Global Footprint Network, 2009, 2019). Demand is the footprint $E = eP$ of a population P with constant per-capita footprint e (gha cap^{-1}), and the shortfall above the human-available amount is

$$S = \left[E - \sigma_f Y_f - \sigma_c Y_c\right]_+, \tag{4}$$

with $[x]_+ = \max(x, 0)$ exact (never smoothed) and $\sigma_f = \sigma_c = 1$ in the baseline. The shortfall is closed by converting capital land into fast land at a finite rate,

$$u_c = \frac{1}{\tau_{\text{conv}}} \min\left(\frac{S}{b_f},\ A_c - A_c^{\min}\right) \qquad [\text{ha yr}^{-1}], \tag{5}$$

where S/b_f is the area that would close the shortfall at the fast-land yield, $A_c^{\min}$ is a typed viability floor, and τ_{conv} is the conversion time constant. The land books evolve by one-to-one area transfers only,

$$\dot{A}_c = -u_c + R_{rc}, \tag{6}$$

$$\dot{A}_f = u_c + \mu A_r - \eta_f A_f, \tag{7}$$

$$\dot{A}_r = -\mu A_r + \eta_f A_f - R_{rc}, \tag{8}$$

where μA_r is cropland maintenance (replanting from the reserve), $\eta_f A_f$ is fast-land retirement to the reserve, and R_{rc} is an explicit, surplus-gated restoration flow from reserve to capital land,

$$R_{rc} = \chi_r\, A_r\, \left[B - E\right]_+, \tag{9}$$

which is the only route back into the capital book. Equations (6)–(8) sum to zero, so the reserve is a constrained balance rather than an independent state. Quality regenerates with a recruitment delay, population follows a delayed logistic law (Pearl, 1925; Hutchinson, 1948) on the carrying capacity $K = B/e$, and degradation acts on the surviving fast-land yield through an

accumulated ecological debt D:

$$\dot{q} = G_c\big(q(t-\tau_g)\big), \tag{10}$$

$$\dot{P} = rP\left(1 - \frac{P}{K(t-\tau_p)}\right), \qquad K = \max(B/e,\, K_{\min}), \tag{11}$$

$$b_f = \big(b_{f0} + T_b(t)\big)\, e^{-\alpha D}, \tag{12}$$

$$\dot{D} = [E - B]_+ - \eta D, \tag{13}$$

$$T_b(t) = \omega_{Tb}\, \frac{\Delta b}{1 + e^{-\kappa_w (t - t_{wave})}}, \qquad \omega_{Tb} \in \{0, 1\}. \tag{14}$$

Because model time t is measured in years and $E - B$ is an annual account amount in gha, the leaky shortfall accumulator D has units gha yr: both $\dot{D}$ and ηD have units gha. This is a model-internal time-integrated liability, not an NFA-reported footprint or biocapacity item.

The technology wave T_b is a bounded logistic yield step controlled by ω_{Tb}: $\omega_{Tb} = 0$ in the baseline, main figures, and all tables except the separately identified matched experiment in SI §S8, where $\omega_{Tb} = 1$. The state is (A_f, A_c, q, P, D) with one delay each in regeneration and demography; a timber-liquidation channel and a direct fast-to-capital restoration flow are stated extensions that remain off in every reported run (Supplementary Information, SI §S1).

Five conventions are essential. (i) *The positive part is exact.* A smoothed ramp would leak conversion when $E < B$; the one-sided deficit mechanism is the model's content, so $[\cdot]_+$ is never regularized and stability at $S = 0$ is one-sided. (ii) *Conversion is a rate.* S/b_f is a stock (ha); it becomes a rate only through τ_{conv}, so u_c is dimensionally ha yr^{-1} and conversion is never instantaneous. (iii) *Area is conserved one-to-one and capacity is a separate book.* Every area flow in (6)–(8) moves hectares between books; regeneration G_c is a flow of capacity into q and never transfers land, so conservation is a property of the equations, not an identity imposed afterwards. (iv) *Annual account amounts and their time-integral are never interchanged.* B, E, Y_f, Y_c and S are annual NFA account amounts in gha; D is a leaky time-integral of shortfall in gha yr, repaid at rate η; b_f, b_c are in gha ha^{-1} and $b_{G,c}$ is a regeneration-value coefficient in gha yr ha^{-1}. (v) *A sustainable cropland equilibrium exists.* Because (7) contains the maintenance term μA_r, a positive fast book A_f^* is sustained without conversion; without it the fast book would decay to zero at every equilibrium.

Two positive feedback loops act on different books. The *conversion loop* reallocates area: $A_c \downarrow\rightarrow A_f \uparrow\rightarrow$ a positive conversion contribution to B (if $b_f > \Upsilon_c$) $\rightarrow K \uparrow\rightarrow P \uparrow\rightarrow S \uparrow\rightarrow A_c \downarrow$; other terms can alter the net change in B, but this loop is the composition mechanism of the paper. The *debt loop* erodes the surviving yield: $D \uparrow\rightarrow b_f \downarrow\rightarrow B \downarrow\rightarrow S \uparrow\rightarrow D \uparrow$; it later turns against the aggregate and is one reason the mask is bounded (Proposition 3).

2.2. Parameters and their status

Table 1 lists the baseline registry. The capital book is a strategic reduction: real ecological capital is a vector of land types with their own yields, values and regeneration times, and the two-book model treats them as one effective class. Consequently most parameters are representative, effective quantities rather than estimates, and the results are stated with that status attached. Two quantities are empirically anchored: the capital-land regeneration timescale, which we bound from raw recovery curves in two ecosystem classes (secondary forest, Poorter et al., 2016; collapsed fish stocks, RAM Legacy, cf. Hutchings and Reynolds, 2004; Neubauer et al., 2013; with soil organic carbon, Poeplau et al., 2011, between them; SI §S5.5), and the provisioning-book yield and area, which we calibrate against NFA cropland biocapacity and the FAO crop-yield index (SI §S5.3). The regeneration anchors bound a *band* of effective timescales (≈25–33 yr, representative ≈29 yr; forest $t_{50} \approx 30$ yr, fishery $t_{50} \approx 5.6$ yr) and are statements about the regeneration rate, not the lag τ_g; the two are kept distinct throughout. The baseline sits in the composition-premium regime, $\Delta b_{\text{conv}} = b_f - \Upsilon_c(q) > 0$, which is the regime in which the mask is possible (Section 3.1); Table 1 therefore fixes a regime, not a prediction.

2.3. Data and the observation operator

The empirical analysis uses the world series of the National Footprint and Biocapacity Accounts (2025 edition): aggregate biocapacity and footprint account values (B, E in gha) for 1961–2022, the land-type biocapacity matrix (cropland, grazing land, forest products, fishing grounds, built-up land), and population (cap). The normalized indices, ratios and log changes used below are dimensionless. The land-type vintage incorporates a retroactive re-basing and gives a slightly higher aggregate growth (+24.5%, $\Delta \ln B = +0.219$) than the aggregate series (+23.0%, $\Delta \ln B = +0.207$); rather than blend the vintages we report the attribution across both (SI §S5.2). The independent proxies that advance past the aggregate are FAOSTAT cropland area (arable

Table 1: Baseline parameter registry (single source of truth for all reported runs) and the status of each parameter. "Representative" values are effective, lumped quantities that are not individually identified; "anchored" values are bounded by independent data (SI §S5); "extension" parameters are stated but off in the base model. Full symbol table with units in SI §S1.

Parameter	Value	Meaning	Status
ρ_c	$0.08\,\mathrm{yr}^{-1}$	quality regeneration rate of capital land	representative; timescale band anchored
b_{f0}	$0.85\,\mathrm{gha\,ha}^{-1}$	base fast-land yield per hectare	representative (NFA cropland book ≈ 0.93 in 1961)
b_c	$0.05\,\mathrm{gha\,ha}^{-1}$	capital-land direct yield	representative
$b_{G,c}$	$0.8\,\mathrm{gha\,yr\,ha}^{-1}$	regeneration-value coefficient for capital land	representative
$q_{\max}$	1.0 (dimensionless)	normalized quality ceiling	normalization
μ	$0.06\,\mathrm{yr}^{-1}$	cropland maintenance (reserve→fast)	representative
η_f	$0.05\,\mathrm{yr}^{-1}$	fast-land retirement (fast→reserve)	representative
τ_{conv}	1.0 yr	conversion time constant	representative
α	$0.03\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	degradation of b_f per unit debt	representative
η	$0.05\,\mathrm{yr}^{-1}$	debt-repayment rate (> 0)	representative
e	$0.55\,\mathrm{gha\,cap}^{-1}$	per-capita footprint (constant)	representative
r	$0.02\,\mathrm{yr}^{-1}$	intrinsic population growth	representative
$\tau_g,\ \tau_p$	20, 25 yr	regeneration and demographic delays	representative; τ_p lumps institutional lags
A_{tot}	2.5 ha	total bioproductive area	normalization
$\sigma_f,\ \sigma_c$	1, 1 (dimensionless)	human-available shares of annual account amounts	baseline (< 1: reservation extension)
χ_r	$0.10\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	surplus-gated restoration coefficient	on with endogenous demand; off under frozen demand
$A_c^{\min}$, $A_r^{\min}$, $K_{\min}$	0.05 ha, 0.10 ha, 0.02 cap	typed floors	domain constraints
Δb, κ_w, t_{wave}	$1.0\,\mathrm{gha\,ha}^{-1}$, $0.1\,\mathrm{yr}^{-1}$, 40 yr	technology wave	off in baseline; active only in matched SI S8 experiment

plus permanent crops; A_f in ha) and a normalized FAO crop-yield index calibrated to the model account yield b_f (gha ha^{-1}) at the base year either by a nominal, dimensionless cropland share ϕ_0 of 1961 biocapacity or directly against measured NFA cropland biocapacity (FAO, 2023); forest area (FAO, 2020), historical land-use reconstructions (Hurtt et al., 2011; Klein Goldewijk et al., 2017), remote-sensing land cover (ESA, 2017; Copernicus, 2019) and cropland-productivity maps (Monfreda et al., 2008) form the hierarchy of extensions described in SI §S5.4.

The decomposition instrument is the exact logarithmic-mean (Divisia) identity. Writing $B_t = X_t + C_t$ with a proxy fast-book component $X_t = \hat{b}_{f,t} A_{f,t}$ and residual $C_t = B_t - X_t$,

$$\begin{aligned} \Delta \ln B &= w_X \big(\Delta \ln b_f + \Delta \ln A_f\big) + w_C \, \Delta \ln C, \\ w_X &= \frac{L(X_T, X_0)}{L(B_T, B_0)}, \qquad w_C = \frac{L(C_T, C_0)}{L(B_T, B_0)}, \end{aligned} \tag{15}$$

with the logarithmic mean $L(a, b) = (a - b)/(\ln a - \ln b)$ for $a \neq b$ and $L(a, a) = a$. The identity holds exactly whenever X and C are positive at both endpoints; average-share weights do not (they overshoot by $\approx 11\%$ here), and the level decomposition $\Delta B = \Delta X + \Delta C$ is the exact fallback when the residual is not positive (SI §S5.2). The extensive/intensive split uses the exact identity $d \ln B = d \ln(B/P) + d \ln P$.

2.4. Numerical solution and verification

The delay system is integrated by the method of steps with an explicit first-order Euler scheme ($\Delta t = 0.05$ yr for the demonstration run, 0.1 and 0.05 yr for the drawdown sweeps). The exact positive part is used; state bounds enforce $A_c \geq A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$, $P, D \geq 0$, and $K \geq K_{\min}$. The constant pre-run histories of B and q cover the maximum active delay; the initial B sample is computed from the initial state, not set to zero. In the SI wave experiment, $T_b(t)$ is applied consistently to the current fast-land yield, returned $B(t)$, and the delayed B history used in $K(t - \tau_p)$; the base-model wave remains off. These v44 corrections remove an artificial time-zero capacity dip and the inconsistent wave bookkeeping identified in the earlier solver. Area conservation is checked at every step ($\max |A_f + A_c + A_r - A_{tot}| = 2.2 \times 10^{-16}$ ha). Every analytic quantity below (balance points, ceiling, eigenvalues, composition diagnostic) is recomputed and asserted before the figure is written. Drawdown thresholds are reported as bands because the operational floor-proximity criterion

depends on tolerance and step size (Section 3.2). The deposited current-quality solver evaluates the regeneration term of (10) at current rather than delayed quality; because q is monotone and the regenerative term is small at baseline ($b_{G,c}G_c \leq 0.016\,\mathrm{gha\,ha^{-1}}$ versus $b_c = 0.05\,\mathrm{gha\,ha^{-1}}$), applying the delay changes the reported quantities by less than 1% (window 149 versus 150 yr; initial-population thresholds within 0.001 cap), and both variants are reported in SI §S3. The new controlled wave run and step-halving checks are in SI §S8. The prior v43 code/data deposit is Zenodo record 22554480; the corrected v44 source and reproduction package is available as described in the Data availability statement.

3. Results

3.1. The composition mask and what the aggregate cannot identify

The per-hectare value of capital land $\Upsilon_c(q)$ is a concave quadratic in q with an interior maximum at $q^* = q_{\max}/2$, where $\Upsilon_c(q^*) = b_c + b_{G,c}\rho_c q_{\max}/4 = 0.066\,\mathrm{gha\,ha^{-1}}$ at the baseline; the regenerative term there ($0.016\,\mathrm{gha\,ha^{-1}}$) is about one third of the direct yield ($0.05\,\mathrm{gha\,ha^{-1}}$; ratio 0.32), so capital value is only modestly enhanced by regeneration. The quantity that governs what conversion does to the aggregate is the *composition diagnostic*

$$\Delta b_{\mathrm{conv}}(q, D) = b_f(D) - \Upsilon_c(q), \tag{16}$$

the difference between what a hectare yields in the fast book and what it is worth in the capital book. At the baseline $\Delta b_{\mathrm{conv}} = +0.787,\ +0.784,\ +0.794\,\mathrm{gha\,ha^{-1}}$ at $q = 0.3, 0.6, 0.9$: the conversion contribution is robustly positive, although other terms still determine the net account change. The three regimes $\Delta b_{\mathrm{conv}} > 0$ (premium: conversion contributes positively to B, so a conversion-driven mask is possible), $\Delta b_{\mathrm{conv}} = 0$ (first-order neutrality, non-generic) and $\Delta b_{\mathrm{conv}} < 0$ (discount: conversion contributes negatively to B, so conversion alone cannot produce that mask) are the only shape distinction the quality-based model admits; because Y_c is linear in A_c at fixed q, there is no area-based capacity maximum and no regime split of the kind an area-based regeneration law would produce.

Proposition 1 (Composition identity, mask and non-identifiability)**.** *Let $T_b \equiv 0$, $R_{rc} \equiv 0$ and hold D fixed, so that b_f is constant. Along any solution of (3)–(10),*

$$\dot{B} = \Delta b_{\mathrm{conv}}(q)\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c}\, G_c'(q)\, A_c\, \dot{q}. \tag{17}$$

(i) *Whenever $u_c > 0$, the conversion contribution to $\dot{B}$ is $\Delta b_{\text{conv}}(q)u_c$ and is positive if and only if $\Delta b_{\text{conv}}(q) > 0$; with $R_{rc} = 0$, the same transfer gives $\dot{A}_c = -u_c < 0$. At any state where the maintenance–retirement imbalance and quality-growth terms vanish ($\mu A_r = \eta_f A_f$ and $\dot{q} = 0$), the net derivative is $\dot{B} = \Delta b_{\text{conv}}(q)u_c$, so its sign has this same condition.*

(ii) *In the reduced frozen-demand, fixed-D case in which $\mu A_r = \eta_f A_f$ and $\dot{q} = 0$ throughout, and with $\sigma_f = \sigma_c = 1$, $\Delta b_{\text{conv}} > 0$ implies $\dot{S} = -\dot{B} = -\Delta b_{\text{conv}} u_c < 0$ whenever $S > 0$. Thus the shortfall decreases to zero if the balance is reached first; if the floor constraint becomes active first, A_c approaches $A_c^{\min}$ asymptotically and the shortfall can persist. This monotonicity is not asserted when the other terms in the full model are nonzero.*

(iii) *At fixed $b_f > 0$ and fixed q (so that $\Upsilon_c(q) > 0$), the observation $B = b_f A_f + \Upsilon_c(q) A_c$ is one scalar constraint on two land areas. At an interior feasible state it leaves a local continuum of decompositions: a sufficiently small δA_c can be offset by $\delta A_f = -(\Upsilon_c/b_f)\delta A_c$, with $\delta A_r = -\delta A_f - \delta A_c$, while preserving B and the area total, provided the area bounds remain satisfied. Thus an aggregate-only observation does not locally identify A_c. For a finite observed change ΔB, the algebraic offset $\Delta A_f = (\Delta B - \Upsilon_c \Delta A_c)/b_f$ is feasible only when the resulting land areas respect their bounds; the admissible range depends on the state and need not be the whole interval from $A_c^{\min} - A_c(0)$ to zero. If b_f and A_f are independently observed, the residual $C = B - b_f A_f = \Upsilon_c(q) A_c$ identifies that product, not Υ_c and A_c separately. An independent observation of A_c would identify the effective value $\Upsilon_c(q) = C/A_c$; even then, b_c, $b_{G,c}$, ρ_c, and q are not separately identified from a single value. With q independently observed over a sufficiently informative range, $\Upsilon_c(q)$ can at most identify b_c and the product $b_{G,c}\rho_c$, not $b_{G,c}$ and ρ_c separately.*

Proof. Differentiate (3) with b_f constant and substitute (6)–(7) with $R_{rc} = 0$: $\dot{B} = b_f(u_c + \mu A_r - \eta_f A_f) - \Upsilon_c u_c + b_{G,c} G_c'(q) A_c \dot{q}$, which is (17). (i) Follows from (17) with $R_{rc} = 0$; when the maintenance–retirement imbalance and quality-growth terms vanish, only the conversion term remains. (ii) Under the reduced-case conditions in the statement, $S = E - B$ and $\dot{S} = -\dot{B} = -\Delta b_{\text{conv}} u_c < 0$ on $S > 0$. Since S is bounded below by zero and A_c is bounded below by $A_c^{\min}$, conversion either closes the shortfall or is limited by

the floor. (iii) At fixed b_f and q, the observation map is linear; an interior feasible point admits the stated small perturbations along a level set of B, while finite changes must also obey the area bounds. The residual C identifies only $\Upsilon_c(q)A_c$ unless A_c is independently observed; the parameter combination follows from the definition of $\Upsilon_c(q)$. □

Figure 1a shows the mask at the baseline under frozen demand $E = 0.825\,\text{gha}$ (restoration off, $A_{f0} = 0.6\,\text{ha}$, $A_{c0} = 1.5\,\text{ha}$, dimensionless $q_0 = 0.9$, $A_r(0) = 0.4\,\text{ha}$). Biocapacity rises from 0.594 to 0.825 gha (+39%) while capital land area A_c falls from 1.500 to 0.808 ha (−46%) and fast land A_f rises from 0.600 to 0.923 ha. The dimensionless quality state q rises from 0.9 to 1.0, but because $q_0 = 0.9 > q^* = q_{\max}/2 = 0.5$, this rise lies on the descending side of the concave per-hectare function $\Upsilon_c(q)$: its value falls from 0.05576 to 0.05000 gha ha^{-1}. Accordingly, the modeled capital contribution $\Upsilon_c(q)A_c$ to B falls from 0.08364 to 0.04038 gha (by 51.7%, about 52%), distinct from the 46% decline in capital-land area. The area changes are one-to-one ($\Delta A_f + \Delta A_c + \Delta A_r = 0$ to machine precision). The dynamics are fast at first and then asymptotic: because the deficit-side land subsystem is affine (Proposition 2(ii)), the approach to the balance point is exponential at the slow rate $|\lambda_1| = 0.054\,\text{yr}^{-1}$ (e-folding ≈ 19 yr), so within 25 yr 98% of the rise in B and 85% of the fall in A_c have occurred ($B = 0.820\,\text{gha}$, $A_c = 0.913\,\text{ha}$; 97% of the fall by 50 yr), while the strict inequalities $\dot{B} > 0$ and $\dot{A}_c < 0$ persist for ≈ 150 yr at numerical resolution. The slow tail is reserve equilibration: retirement of fast land to the reserve lowers the fast flow, re-opens a small shortfall, and converts a little more capital, until the reserve reaches its equilibrium share $A_r/A_f = \eta_f/\mu$. The window is therefore not a fixed horizon: its scale is set by λ_1, which depends on τ_{conv}, μ, η_f and the yield ratio b_c/b_f (SI §S2.2), and its amplitude by Δb_{conv} and the initial deficit.

Two consequences of Proposition 1(iii) frame the empirical analysis of Section 3.5. First, the composition signature (B rises while A_c falls) is contemporaneously observable only in the two-book series; a monitor who sees the aggregate alone cannot distinguish a composition mask from broad growth. Second, independent proxies for the fast book (A_f, b_f) convert the aggregate into a measured fast book and a residual, which is the most that can be identified without independent ecological valuation of the capital book. We call the net two-book signature B rising while A_c falls the *composition mask*; its conversion component is positive when $\Delta b_{\text{conv}} > 0$, but the net sign of $\dot{B}$

also depends on the other terms in (17). The alternative *liquidation mask* is E steady while A_c falls, which does not require a positive conversion contribution. The two-book series makes these readings distinguishable.

3.2. The drawdown is a demand-driven floor collision, not a fold

With endogenous population there is no interior steady state with active conversion: at a demographic equilibrium $P = K = B/e$, so $E = B$ and $S = 0$. Conversion is therefore a transient regime, and its steady states exist only under a frozen exogenous demand pinned above the flow. The frozen-demand system is where the question of a catastrophic shift must be posed, and it has a complete answer.

Proposition 2 (Balance point, demand ceiling and affine deficit-side dynamics)**.** *Assume frozen demand E, $T_b \equiv 0$, $R_{rc} \equiv 0$, $q = q_{\max}$ (the attractor of (10) for $q(0) > 0$) and $D = 0$. Let $k = \mu/(\mu + \eta_f)$ and*

$$B_{eq}(A_c) = b_{f0}\,k\,(A_{tot} - A_c) + b_c A_c = b_{f0} k A_{tot} - (b_{f0}k - b_c)A_c. \tag{18}$$

(i) Equilibria of the land subsystem on the no-conversion face $S = 0$ satisfy $A_f = k(A_{tot} - A_c)$, $A_r = (1-k)(A_{tot} - A_c)$ and $B_{eq}(A_c) \geq E$. If $b_{f0}k > b_c$, B_{eq} is strictly decreasing, so the balance point $B_{eq}(A_c^*) = E$ is unique,

$$A_c^*(E) = \frac{b_{f0}kA_{tot} - E}{b_{f0}k - b_c}, \tag{19}$$

The algebraic root lies in the feasible land interval $A_c \in [A_c^{\min}, A_{tot}]$ exactly when $B_{eq}(A_{tot}) \leq E \leq E_{\text{ceil}} := B_{eq}(A_c^{\min})$; it lies strictly above $A_c^{\min}$ iff $E < E_{\text{ceil}}$. For $E < B_{eq}(A_{tot})$, the root lies above A_{tot} and feasible no-conversion equilibria already satisfy $B_{eq} \geq E$.

(ii) On the deficit side ($S > 0$, $S/b_f < A_c - A_c^{\min}$) the land subsystem is affine in $z = (A_f, A_c)$, $\dot{z} = Jz + c$, with

$$J = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix}, \qquad \det J = \frac{\mu + \eta_f}{\tau_{\text{conv}}}\Big(k - \frac{b_c}{b_f}\Big), \tag{20}$$

and $\operatorname{tr} J < 0$. J does not depend on E or on the state, $\det J > 0$ exactly when B_{eq} is decreasing, and the fixed point of the affine system

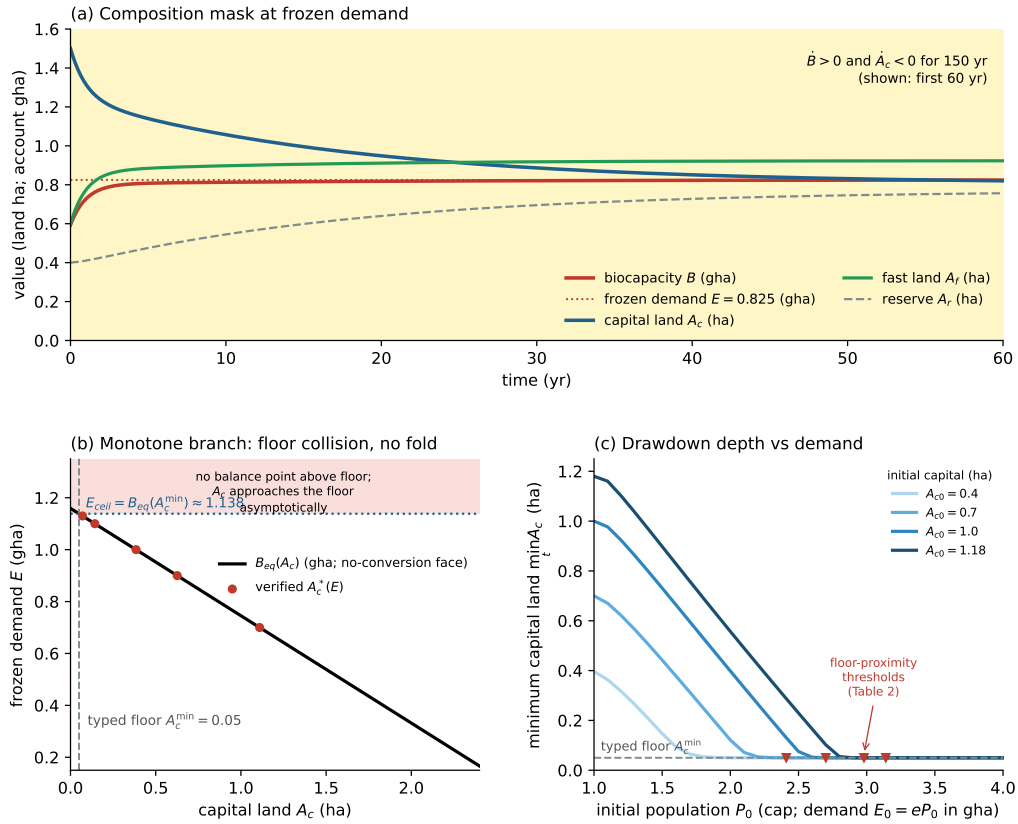


Figure 1: Model dynamics at the baseline registry (Table 1), regenerated from the deposited code. Account amounts B and E are in gha; land areas A_f, A_c, A_r are in ha; population P_0 is in cap. (a) Composition mask under frozen demand $E = 0.825$ (restoration off, $A_{f0} = 0.6$, $A_{c0} = 1.5$, dimensionless $q_0 = 0.9$): biocapacity B rises from 0.594 to 0.825 while capital land A_c falls from 1.500 to 0.808 and fast land A_f rises from 0.600 to 0.923; the shaded region marks the interval on which $\dot{B} > 0$ and $\dot{A}_c < 0$ hold simultaneously (≈ 150 yr at numerical resolution; only the first 60 yr are shown; 98% of the rise in B and 85% of the fall in A_c occur within the first 25 yr). (b) The frozen-demand balance branch $B_{eq}(A_c)$ of Proposition 2, strictly decreasing in A_c, with the verified balance points $A_c^*(E) = 1.110, 0.626, 0.385, 0.143, 0.070$ at $E = 0.70, 0.90, 1.00, 1.10, 1.13$; the branch meets the typed floor $A_c^{\min} = 0.05$ at the demand ceiling $E_{\text{ceil}} \approx 1.138$, above which no admissible balance point exists and the capital book approaches the floor (shaded). (c) Endogenous-demand runs ($\tau_g = 20$, $\tau_p = 25$ yr, $\chi_r = 0.10$): the minimum capital land reached during a run is a continuous, monotone function of initial demand P_0 (hence $E_0 = eP_0$) for each initial capital A_{c0}; the floor is approached asymptotically, and the floor-proximity thresholds of Table 2 (triangles) are the operational points at which the floor is approached to within 10^{-6} ha. There is no discontinuity in the outcome as a function of demand.

is the balance point of (i). For $B_{eq}(A_{tot}) \le E < E_{\mathrm{ceil}}$ the balance point lies above the typed floor and every trajectory that remains in this regime converges to it at the constant rates given by the eigenvalues of J; for $E > E_{\mathrm{ceil}}$ the affine fixed point lies below the floor, so the trajectory approaches the face $A_c = A_c^{\min}$ asymptotically, with a limiting persistent shortfall $S^* = E - E_{\mathrm{ceil}} > 0$. The ceiling is a collision of the equilibrium with a face of the admissible set, not a bifurcation of the vector field: there is no second branch, no fold, no hysteresis, and the leading eigenvalue does not tend to zero as $E \to E_{\mathrm{ceil}}$.

(iii) Beyond the ceiling, with degradation active, the debt converges to the unique root $D^*(E)$ of $\eta D = E - B_{eq}(A_c^{\min}; D)$, where $B_{eq}(\cdot\,; D)$ uses $b_{f0}e^{-\alpha D}$, provided $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min})$; D^* is increasing in E.

Proof. (i) With $S = 0$ and $R_{rc} = 0$, (7) gives $\mu A_r = \eta_f A_f$, hence $A_r/A_f = \eta_f/\mu$ and, with $A_r = A_{tot} - A_f - A_c$, $A_f = k(A_{tot} - A_c)$; substituting into (3) with $q = q_{\max}$ ($\Upsilon_c = b_c$) and $D = 0$ gives $B = B_{eq}(A_c)$, and $S = 0$ requires $B \ge E$. Strict monotonicity and (19) follow from $b_{f0}k > b_c$. (ii) On the deficit side $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\mathrm{conv}})$ is affine in z; substituting into (6)–(7) with $A_r = A_{tot} - A_f - A_c$ gives J and a constant vector c by direct differentiation; $\det J$ follows by expansion, and $\operatorname{tr} J = -\tau_{\mathrm{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f < 0$ for $b_c < b_f$. The fixed point solves $u_c = 0$ and $\mu A_r = \eta_f A_f$, i.e. it is the balance point. Because J is constant, convergence rates are constant along the branch. (iii) The map $D \mapsto \eta D + B_{eq}(A_c^{\min}; D) - E$ is continuous, negative at $D = 0$ when $E > E_{\mathrm{ceil}}$, tends to $+\infty$, and is strictly increasing when $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min}) \ge -\partial_D B_{eq}$; hence it has a unique root, which increases with E. □

At the baseline $k = 0.5455$, the slope $b_{f0}k - b_c = 0.414\,\mathrm{gha\,ha^{-1}}$, and $E_{\mathrm{ceil}} = 1.138\,\mathrm{gha}$; the balance points $A_c^*(E) = 1.110, 0.626, 0.385, 0.143, 0.070\,\mathrm{ha}$ at $E = 0.70, 0.90, 1.00, 1.10, 1.13\,\mathrm{gha}$ agree with the integrated equilibria (Fig. 1b). The eigenvalues of J are real and negative, $\lambda_1 = -0.0537\,\mathrm{yr^{-1}}$ and $\lambda_2 = -0.998\,\mathrm{yr^{-1}}$; λ_1 varies only weakly with the fast-land yield, from $-0.0522\,\mathrm{yr^{-1}}$ at $b_f = 0.60$ to $-0.0556\,\mathrm{yr^{-1}}$ at $b_f = 2.00$ (Table 2a), and never changes sign. There is no universal scalar stability index: the conversion-loop gain is the state- and parameter-dependent Δb_{conv}, of order $1\,\mathrm{gha\,ha^{-1}}$, and the return rate is $|\lambda_1|$ of order $10^{-2}\,\mathrm{yr^{-1}}$. Beyond the ceiling the debt fixed points are $D^* = 1.19$ and $5.38\,\mathrm{gha\,yr}$ at $E = 1.158$ and $1.238\,\mathrm{gha}$ (integrated values

1.21 and 5.39 gha yr); the monotonicity condition of (iii) holds with margin ($\alpha b_{f0} k(A_{tot} - A_c^{\min}) = 0.034\,\mathrm{yr}^{-1} < \eta = 0.05\,\mathrm{yr}^{-1}$), so under frozen demand the debt saturates rather than running away.

The statement "no critical slowing down" is precise here. By critical slowing down we mean that the least-negative eigenvalue of the relevant linearization tends to 0^- as a parameter approaches a fold, so that the return time $T_{ret} \approx -1/\lambda_1$ diverges. In Proposition 2 the linearization is constant along the entire branch, so neither λ_1 nor $T_{ret} \approx 19\,\mathrm{yr}$ changes as $E \to E_{\mathrm{ceil}}$. The only early-warning signal the model offers is a *boundary* signal, the capital book approaching its typed floor, which is observable only in the two-book series.

With endogenous demand the same structure appears as a continuous drawdown. Figure 1c shows the minimum capital land reached during a run as a function of initial population P_0 for four initial capital-land areas A_{c0}: the curves are continuous and monotone, the floor is approached asymptotically (because $u_c \leq (A_c - A_c^{\min})/\tau_{\mathrm{conv}}$, the distance to the floor decays exponentially during a sustained deficit and is never crossed), and higher initial capital-land area supports higher demand before the floor is approached. The "collapse thresholds" of the earlier literature on this model are therefore operational quantities: Table 2b reports the initial demand at which the floor is approached to within 10^{-2}, 10^{-4} and 10^{-6} ha of its value. The 10^{-6} ha floor-proximity thresholds ($P_0 \approx 2.41$, 2.70, 2.98, 3.14 cap for $A_{c0} = 0.40$, 0.70, 1.00, 1.18 ha) are obtained by bisection on that tolerance criterion; they rise monotonically with initial capital, as do the looser criteria, and no criterion exhibits a jump. Two robustness checks bear on the predictions of Section 3.6: the initial-population floor-proximity threshold at $A_{c0} = 1.0$ ha is identical to three decimals for $\tau_g = 5$, 20 and 40 yr, and identical to three decimals with the restoration coefficient $\chi_r = 0.10\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$ and $\chi_r = 0\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$ (SI §S4). The first reflects the small leverage of the regenerative term at the baseline and the monotone quality dynamics; the second reflects the fact that a surplus-gated flow is inert during the shortfall that produces the drawdown.

3.3. The mask is bounded: technology gain is parameter-bounded, debt erosion is shortfall-bounded

Both the technology wave and the degradation act on the same coefficient, the fast-land yield b_f of (12), and they do so multiplicatively. This yields a structural asymmetry.

Table 2: Model results at the baseline registry. (a) Composition diagnostic at $q^* = q_{\max}/2$ ($\Upsilon_c(q^*) = 0.066\,\mathrm{gha\,ha^{-1}}$), eigenvalues of the deficit-side matrix J of (20) and the demand ceiling, for a sweep of the fast-land yield b_f (all other parameters at baseline). (b) Initial population P_0 (cap; initial footprint $E_0 = 0.55P_0$ in gha) at which the minimum capital land reached during an endogenous-demand run ($A_{f0} = 0.6\,\mathrm{ha}$, dimensionless $q_0 = 0.9$, $\tau_g = 20\,\mathrm{yr}$, $\tau_p = 25\,\mathrm{yr}$) comes within the stated distance of the typed floor $A_c^{\min} = 0.05$; the first two columns are read from a P_0 grid of step 0.1 ($\Delta t = 0.1\,\mathrm{yr}$, $T = 400\,\mathrm{yr}$), the last is a bisection using the $10^{-6}\,\mathrm{ha}$ floor-proximity criterion ($\Delta t = 0.05\,\mathrm{yr}$, $T = 1000\,\mathrm{yr}$). Drawdown depth is continuous in P_0 (Fig. 1c); the thresholds are a band, not a single value.

(a) Fast-land yield sweep (all rows composition-premium, $\Delta b_{\mathrm{conv}} > 0$)

b_f (gha ha^{-1})	$\Delta b_{\mathrm{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.804
0.85 (baseline)	+0.784	−0.0537	−0.998	1.138
1.20	+1.134	−0.0547	−1.014	1.606
2.00	+1.934	−0.0556	−1.029	2.675

(b) Initial demand P_0 at which $\min_t A_c$ approaches $A_c^{\min}$ to within:

A_{c0} (ha)	within 10^{-2} ha	within 10^{-4} ha	within 10^{-6} ha
0.40	≈ 1.8	≈ 2.1	≈ 2.41
0.70	≈ 2.2	≈ 2.4	≈ 2.70
1.00	≈ 2.6	≈ 2.8	≈ 2.98
1.18	≈ 2.8	≈ 3.0	≈ 3.14

Proposition 3 (Boundedness asymmetry)**.** *Under* (12)–(14)*:*

(i) the technology factor satisfies $1 \le (b_{f0}+T_b(t))/b_{f0} \le 1+\Delta b/b_{f0}$ *for all* t*, so the logarithmic gain of the yield channel is bounded by* $\ln(1+\Delta b/b_{f0})$ *uniformly in time (*0.78 *at the baseline);*

(ii) the debt satisfies $D(t) \le \max\{D(0),\ \sup_{s\le t}[E-B]_+(s)/\eta\}$*, and* $D \to S^*/\eta$ *whenever the shortfall settles at* S^**; the degradation factor* $e^{-\alpha D}$ *is therefore bounded below only through the sustained shortfall;*

(iii) the degradation loss exceeds the maximal technology gain once the sustained shortfall exceeds $S^\dagger = (\eta/\alpha)\ln(1+\Delta b/b_{f0})$*, equal to* 1.30 gha *at the baseline.*

Under an additive degradation law $b_f = b_{f0} + T_b - \alpha D$ *the two channels enter symmetrically and (iii) reduces to a comparison of* Δb *with* αD^**; the asymmetry is a property of the multiplicative form.*

Proof. (i) is $0 \le T_b \le \Delta b$. (ii) From (13), $\dot D \le \bar S - \eta D$ with $\bar S = \sup_{s\le t}[E-B]_+$, so $D(t) \le \max\{D(0), \bar S/\eta\}$ by comparison; if $[E-B]_+ \to S^*$ then $D \to S^*/\eta$. (iii) compares $\alpha S^*/\eta$ with $\ln(1+\Delta b/b_{f0})$. □

The proposition corrects a tempting overstatement: debt does not compound without bound under a fixed shortfall – it saturates at S^*/η – but its ceiling is set by the shortfall rather than by a parameter, and the shortfall is itself raised by demand growth. At the baseline the crossover shortfall $S^\dagger \approx 1.3$ gha exceeds the baseline biocapacity, so at these parameters the debt channel is a slow erosion rather than a reckoning; what bounds the mask at the baseline is the balance-point mechanism of Proposition 1(ii), with debt erosion as the second, slower limit. A second coupling is possible: because $K = B/e$, a technology-driven rise in B can raise carrying capacity and, after the demographic delay, demand. Whether this produces a larger all-time debt peak depends on the initial state and the timing of the response; it is not a sign-general theorem. In the controlled matched experiment in SI §S8, with $P_0 = 1.0$ cap, the active T_b run reaches a global D maximum of 0.02020 gha yr versus 0.00813 gha yr without the wave (ratio 2.49). At the reference $P_0 = 1.2$ cap, the all-time peak is instead lower with the wave (0.05104 versus 0.06122 gha yr), while the maximum over $t \ge t_{wave} = 40$ yr is higher (0.02518 versus 0.01085 gha yr; ratio 2.32). This delayed-rebound response is conditional on the declared parameters and initial demand, and

the global maximum must not be conflated with the post-wave-center maximum. The active-wave experiment is a diagnostic extension, not a forecast or an empirical estimate.

3.4. Recovery requires an explicit, gated restoration flow

The model is structurally asymmetric: $A_c \to A_f$ proceeds through conversion, but no term in (6)–(8) returns land to the capital book except the restoration flow R_{rc}. Fast-land retirement sends land to the reserve, not to capital. Reducing population, footprint or shortfall therefore does not reverse prior conversion: with $\chi_r = 0$ the capital book stays where the shortfall left it. In the endogenous-demand run with $A_{c0} = 1.0\,\mathrm{ha}$ and $P_0 = 2.0\,\mathrm{cap}$, capital land falls to 0.398 ha within four years and remains there without restoration, whereas with the surplus-gated flow ($\chi_r = 0.10\,\mathrm{gha^{-1}\,yr^{-1}}$) it recovers to 0.828 ha, a dimensionless recovery fraction $\mathcal{R}_c = (A_c(T) - A_c^{\mathrm{deg}})/(A_c^{\mathrm{init}} - A_c^{\mathrm{deg}}) = 0.71$ (SI §S4.3). The sign of the gate matters. A surplus-gated flow, active when $B > E$, is the ecologically sign-correct rewilding: it retires land only when the system has spare capacity, but it is inert during the shortfall it must eventually address – which is why the drawdown thresholds of Table 2b do not depend on χ_r – and once a surplus opens it can over-restore while the supported population re-equilibrates downward, so a target-level stopping rule is needed for a clean $\mathcal{R}_c = 1$ (SI §S4.4). A deficit-gated flow, active when $E > B$, is the one gate that acts during the shortfall, but it retires productive land exactly when provision is short and can deepen the near-term deficit; at the tested rates conversion outpaces it, so it is necessary but insufficient and must be sized or capped. Neither gate is an unqualified good, and the gate-sign asymmetry is phase-dependent rather than a universal ranking.

3.5. Measured composition of aggregate biocapacity, 1961–2022

The NFA world series records annual account values B, E in gha and population P in cap; the per-capita levels $B/P, E/P$ are in $\mathrm{gha\,cap^{-1}}$. Indices, ratios and logarithmic changes below are dimensionless. The identity $d\ln B = d\ln(B/P) + d\ln P$ applied to the NFA world series gives the aggregation face of the result (Fig. 2). Aggregate biocapacity rose +23.0% ($\Delta\ln B = +0.207$, $+0.339\%\,\mathrm{yr^{-1}}$); per-capita biocapacity fell −52.9% ($\Delta\ln(B/P) = -0.752$, $-1.23\%\,\mathrm{yr^{-1}}$); the per-capita footprint rose +16.9%; and the biocapacity ratio $R_B = E/B$ rose from 0.718 to 1.782, crossing 1 in 1971. Because the identity is exact (dimensionless residual $< 10^{-16}$) the split is unambiguous:

population contributed +0.959 (+464% of $\Delta \ln B$) and the per-capita term −0.752 (−364%). This is the extensive/intensive split, not a causal claim (biocapacity is computed from area and yield independently of P), and it is not yet the two-land composition mask: the same aggregate that hides a per-capita halving is the one that would hide a capital-land drawdown.

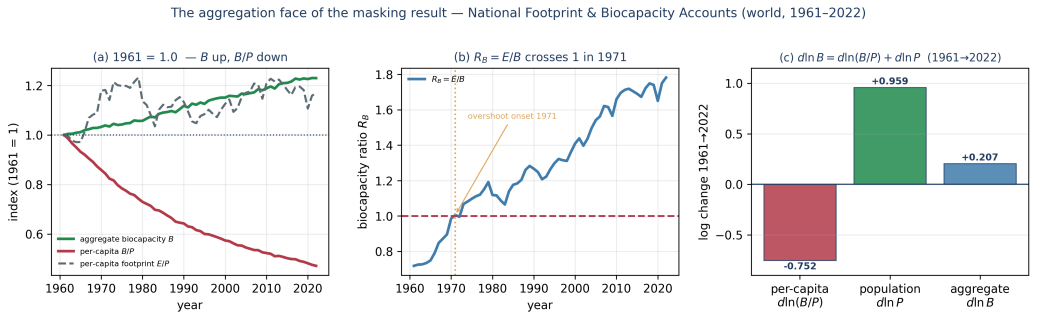


Figure 2: The aggregation face of the masking result, National Footprint and Biocapacity Accounts (world, 1961–2022). The underlying annual account values B, E are in gha and population P is in cap; per-capita levels $B/P, E/P$ are in gha cap^{-1}. Plotted indices (including the per-capita indices), ratios and log contributions are dimensionless. (a) Indices (1961 = 1) of aggregate biocapacity B, per-capita biocapacity B/P and per-capita footprint E/P: B rises while B/P halves. (b) The biocapacity ratio $R_B = E/B$ crosses 1 (overshoot onset) in 1971. (c) The exact identity $d\ln B = d\ln(B/P) + d\ln P$: population +0.959 and per-capita −0.752 (shares of $\Delta \ln B$: +464% and −364%). The public aggregate carries no yield/area split, so this panel is explicitly not the two-land composition mask.

To advance past the aggregate we apply the observation operator of Section 2.3. With A_f from FAOSTAT cropland area and b_f from the normalized crop-yield index calibrated at 1961, the exact Divisia decomposition (15) attributes the +0.207 aggregate log-change to a yield channel +0.393, a cropland-area channel +0.056 and a residual −0.242 (Table 3). Raw component log-changes (+1.128, +0.160, −0.404) are changes in the components, not contributions, and cannot be summed; the log-mean weights $w_X = 0.349$, $w_C = 0.600$ make the identity exact, whereas average shares (0.375, 0.625) would give +0.231. The yield channel dominates the growth of the aggregate, cropland area is a small positive, and the residual – all biocapacity not captured by the cropland proxy – is a large negative. The result is robust to the unidentified base-year cropland share: over the dimensionless share $\phi_0 \in [0.10, 0.34]$ the weighted identity sums to +0.207 exactly and the yield channel is the dominant positive term at every ϕ_0,

although the split is not pinned (w_X ranges 0.18–0.62); for $\phi_0 \geq 0.339$ the residual is no longer positive at the endpoint and the level decomposition $\Delta B = \Delta X + \Delta C$ (reproducing $\Delta B = +2.242 \times 10^9$ gha) carries the same sign statement (SI §S5.2). The nominal anchor $\phi_0 = 0.19$ can be replaced by the measured one, $b_f(1961) = B_{\text{cropland,NFA}}(1961)/A_f(1961) \approx 0.93\,\text{gha}\,\text{ha}^{-1}$ (cropland share ≈ 0.128), which lies inside the robustness band and leaves the decomposition unchanged. The instantaneous fast-book share s_X rises from 0.19 to 0.56 over the period; only the growth and sign statements are robust, and b_c, $b_{G,c}$, ρ_c and the yield/area split of the residual remain unidentified, exactly as Proposition 1(iii) says they must.

Table 3: Exact log-mean (Divisia) decomposition of world aggregate biocapacity growth, 1961–2022 (NFA aggregate series; cropland proxies from FAOSTAT; base-year calibration $\phi_0 = 0.19$). Account values B, X, C are in gha; log changes, weights and contributions are dimensionless. Contributions sum exactly to $\Delta \ln B = +0.207$. Raw log-changes are component changes, not contributions.

Component	Raw $\Delta \ln$	Log-mean weight	Contribution to $\Delta \ln B$
Fast-book yield b_f (crop-yield index)	+1.128	$w_X = 0.349$	+0.393
Fast-book area A_f (cropland)	+0.160	$w_X = 0.349$	+0.056
Residual $C = B - b_f A_f$ (non-cropland)	−0.404	$w_C = 0.600$	−0.242
Total			+0.207

The measured land-type accounts make the composition explicit without any proxy (Fig. 3). In the land-type vintage the aggregate grew +24.5% ($\Delta \ln B = +0.219$), of which cropland contributed +23.8 points and built-up land +7.3 points, while forest products, grazing land and fishing grounds together contributed −6.6 points; the component ratios are cropland 2.85× (area 1.17×, per-hectare yield 2.43×), built-up 8.63×, forest products 0.90×, grazing 0.98×, fishing 0.96×. Cropland accounts for essentially all of the net aggregate growth ($\approx$ 97–103% depending on vintage), and the non-cropland book is flat at the endpoint (1.008×; intra-period index up to 1.024×). The identifiability limit is recursive: the near-flat non-cropland total is itself the net of declining forest-products, grazing and fishing biocapacity (together $\approx -6.4 \times 10^8$ gha) and an 8.6× rise in built-up land. These categories are accounting flows and land classes, not direct measurements of the model's

effective area state A_c or its q-dependent contribution $\Upsilon_c(q)A_c$; they do not demonstrate a measured decline in A_c, but they do show that aggregate summary statistics conceal composition at more than one level.

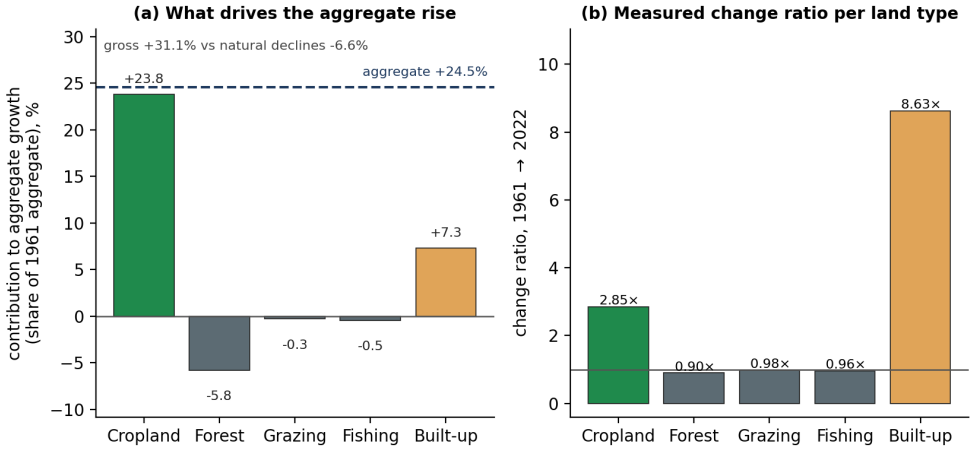


Figure 3: Measured land-type composition of aggregate biocapacity change, National Footprint and Biocapacity Accounts (world, 1961–2022, land-type vintage). The underlying annual biocapacity account values are in gha; plotted contributions are percentage points and endpoint ratios are dimensionless. (a) Contributions to the +24.5% aggregate growth ($\Delta \ln B = +0.219$): cropland +23.8 points and built-up land +7.3 points (gross +31.1), against forest products, grazing and fishing −6.6 points; the components sum exactly to the aggregate. (b) Endpoint ratios: cropland 2.85×, built-up 8.63×, forest products 0.90×, grazing 0.98×, fishing 0.96×. A single aggregate number cannot reveal this composition.

3.6. Falsifiable predictions

The model's testable content, framed against two-book series (B, A_c, A_f and land-cover/yield proxies) rather than the aggregate, is the following.

P1. **Mask signature.** In the baseline frozen-demand run, the model predicts a finite interval with $\dot{A}_c < 0$ and $\dot{B} > 0$ (Fig. 1a). The conversion contribution is positive exactly when $\Delta b_{\text{conv}} > 0$, but the net sign of $\dot{B}$ also depends on maintenance/retirement, quality growth, debt, technology, and restoration. With baseline shares $\sigma_f = \sigma_c = 1$, active conversion requires $R_B = E/B > 1$; in the reserved-flow extension with $\sigma_f, \sigma_c < 1$, conversion can occur even when $R_B \leq 1$, which is not a baseline result. A two-book series distinguishes the net composition signature (B rises while A_c falls) from a liquidation mask (E steady while A_c falls) (Proposition 1).

P2. **Window scale.** The duration of the mask is set by the slow eigenvalue λ_1 (through τ_{conv}, μ, η_f, b_c/b_f) and its amplitude by Δb_{conv} and the deficit, not by a fixed horizon; the bulk of conversion precedes the closing of the deficit by one to two e-folding times.

P3. **No critical-slowing precursor.** The return time does not grow as demand approaches the ceiling, and the drawdown depth is continuous in demand (Fig. 1c); the observable warning is the capital book approaching its typed floor (Proposition 2). This is a negative prediction: a catastrophic-shift early-warning indicator is not expected to precede the two-land drawdown.

P4. **Demand threshold, not delay threshold.** The operational initial-demand threshold for approaching the capital floor rises with initial capital (Table 2b) and is insensitive to the regeneration delay (identical to 10^{-3} for $\tau_g = 5$–$40\,\text{yr}$), because the regenerative share of capital value is small and quality is monotone; a one-stock regeneration cliff is not expected.

P5. **Recovery requires a gated flow.** Reducing demand does not restore the capital book; recovery requires an explicit restoration flow whose gate sign determines whether it is inert during the shortfall (surplus-gated; thresholds unchanged by χ_r) or deepens it (deficit-gated).

The identifiability statement of Proposition 1(iii) is not a prediction but a constraint on testing: P1–P5 can be tested only with independent A_c and b_f proxies, never with the aggregate alone.

4. Discussion

4.1. What the two-land structure changes, and what it removes

The results depart from the single-stock, single-aggregate reading of the environment that dominates the resource and sustainability literature (Schaefer, 1954; May, 1973; Meadows et al., 1972, 2004; Wackernagel and Rees, 1996; Brander and Taylor, 1998) in three respects, each of which is a negative result as much as a positive one. First, the conversion component can raise aggregate capacity while capital-land area A_c falls, but this bounded composition effect does not by itself determine the net change in the aggregate: Proposition 1 gives the exact sign condition for the conversion contribution, and Proposition 3 explains why the technology/debt mask cannot

persist. Second, the drawdown of capital-land area A_c is a demand-driven floor collision. The alternative-stable-states literature (Scheffer et al., 2001) predicts that a system approaching a fold shows critical slowing down, hysteresis and a discontinuous jump; none of these is present here, and their absence is structural (Table 4): the deficit-side dynamics are affine with constant real eigenvalues, the equilibrium leaves the admissible set through a face rather than through a bifurcation, and the drawdown depth is continuous in demand. Third, recovery is not the reverse of degradation; it requires an explicit, gated flow, and the gate sign is a design decision with phase-dependent consequences.

The negative results have a direct bearing on early-warning practice. Generic indicators of critical slowing down (rising autocorrelation and variance) are motivated by the approach to a fold (Scheffer et al., 2001). In a two-book system in which the drawdown is a boundary collision, such indicators would not be expected to fire, and their silence should not be read as safety. The informative signal is the capital book itself, which requires the two-book series.

Several quantities that circulate for models of this kind are deliberately not claimed here. The two-gain Hopf classification of which delay destabilizes, the associated critical delays, and the reference eigenvalue of order $+0.6\,\mathrm{yr}^{-1}$ belong to a single-stock fast–slow reduction that we retain only as a comparator in SI §S7, where its equations, equilibrium family and figures are given; none of its features is carried into the two-land results. The "18–20 yr regeneration cliff" of the one-stock comparator does not survive the two-land structure (Table 4). What is structural in the two-land model is the composition logic and the identifiability limit; what is conditional on effective parameters is the sign and size of Δb_{conv} and the location of the ceiling.

4.2. Measurement implications

The methodological result of the paper is the identifiability limit. Biocapacity in global hectares is a value-weighted scalar, not a count of the underlying assets. Given only that a portfolio's value rose by 20%, one cannot tell whether every holding grew a little or some made large gains while others quietly fell; the scalar gloss hides the composition. This is the generic limit of any value-weighted composite (Nardo et al., 2008), and Proposition 1(iii) states it for biocapacity: the aggregate identifies one linear combination of the areas, and the capital-book parameters enter only through identifiable

Table 4: Structural properties and negative results in the two-land model, each derived under the two-land structure rather than inherited from a one-stock comparator.

Phenomenon	Present?	Status in the two-land model
Capital-land catastrophic fold	No	$B_{eq}(A_c)$ is strictly monotone; the balance point is unique; the transition is a typed-floor collision at $E_{\text{ceil}} \approx 1.138\,\text{gha}$ (Proposition 2)
Critical slowing down	No	deficit-side dynamics are affine; $\lambda_1 \approx -0.054\,\text{yr}^{-1}$ is constant along the branch and does not tend to 0
Hysteresis / path dependence of the balance point	No	no second branch; irreversibility of the *area* book is a different property (no return flow without R_{rc}, Section 3.4)
Discontinuous transition with demand	No	drawdown depth is continuous and monotone in initial population P_0 (cap; Fig. 1c); thresholds are tolerance-dependent bands
Regeneration-delay cliff	No	operational initial-population floor-proximity threshold identical to three decimal places for $\tau_g = 5$–$40\,\text{yr}$; the regenerative share of capital value is small and q is monotone
Sustained boom–bust cycle	No	single overshoot pulse; q is monotone; $r\tau_p = 0.5 < \pi/2$, below the Hutchinson threshold of the delayed logistic population loop (Hutchinson, 1948)
Allee-type rescue threshold in capital	No	no capital level below which recovery is impossible other than the floor itself; the lever is demand or restoration, not the lag

combinations. The aggregate is therefore a weak-sustainability index in the precise sense of the companion assessment work (Abaee, 2026b): it is the linear, perfectly substitutable member of the aggregator family (its Lemma A), and it can satisfy an aggregate floor while the typed floor on A_c is violated by exactly the amount that the fast-land yield gains against the capital loss. In that family no positive-elasticity aggregator is uniformly safe and only the typed, non-compensating one is (its Theorem S2(ii)). Under exact-tube semantics, in which a transition is safe only if every state along it satisfies the constraints (Abaee, 2026b), the passage during which A_c falls is itself a violation even if the endpoint recovers – a reason to report through-time trajectories and boundary curves, not endpoint attractors.

For the National Footprint Accounts the implication is concrete. The accounts are account-based and do not capture soil erosion, deforestation, groundwater depletion or other cumulative degradation, so estimated biocapacity likely overstates the real base and the footprint understates the overshoot (Wackernagel et al., 2002; Borucke et al., 2013); critics have made the point forcefully (Blomqvist et al., 2013; Giampietro and Saltelli, 2014; Galli et al., 2016; van den Bergh and Grazi, 2015). The present results add a different and more specific point: even an internally consistent aggregate cannot be inverted into its components. A growth headline for total biocapacity carries no information about whether cropland growth is compensating for stagnation or offsetting declines elsewhere, and the measured accounts show that this is exactly what happened in 1961–2022 (Fig. 3). The informative assessment reports components with the total – at minimum component endpoint ratios, log contributions, attribution shares, data vintage and endpoint sensitivity – at every level at which an aggregate is used. Parallel observations have been made for human appropriation of net primary production and for land-management data (Haberl et al., 2007; Krausmann et al., 2013; Erb et al., 2017; Carvalhais et al., 2014): the composition of an aggregate account value, not its level, carries the sustainability information.

The model's capital decline has a different evidential status from those account movements. In the frozen-demand baseline (Fig. 1a), modeled capital land A_c falls from 1.500 to 0.808 ha while B rises from 0.594 to 0.825 gha; because restoration is off in that run, the reduced land allocation persists after the shortfall closes. The quality state q rises from 0.9 to 1.0, but since $q_0 = 0.9 > q^* = q_{\max}/2 = 0.5$, this rise lies on the descending side of the concave per-hectare function $\Upsilon_c(q)$. The logistic regeneration term falls, so $\Upsilon_c(q)$ declines from 0.05576 to 0.05000 gha ha^{-1}; the effective capital ac-

count, i.e. the modeled capital contribution $\Upsilon_c(q)A_c$ to B, consequently falls from 0.08364 to 0.04038 gha (by 51.7%, about 52%), distinct from the 46.2% decline in capital-land area. This is a conditional model trajectory, not an estimate of a global physical-stock trend. NFA's forest-products, grazing and fishing categories are annual account values built from land classes, yields and conversion factors. Their modest declines are relevant composition signals, but they do not identify ΔA_c, biomass, soil condition or any other physical capital-stock change.

A defensible physical-stock scoreboard should therefore preserve a vector of component accounts rather than add unlike stocks into a single unweighted total. Report ecosystem extent (for example, forest or wetland area in ha) separately from condition or stock measures (for example, forest growing stock or biomass in m^3 or tonnes of carbon, soil organic carbon in tonnes of carbon for a stated depth, fish spawning-stock biomass in tonnes, and freshwater storage in m^3). Keep regenerative and appropriation flows separate from stocks: net annual increment, soil-carbon accumulation or loss, fish recruitment and catch, and water recharge and withdrawal are not interchangeable with the corresponding stock levels. For each component, provide the reference year, spatial boundary, physical unit, absolute change and baseline ratio, uncertainty, data vintage, and any relevant ecological floor or reference range. Do not double-count nested measures such as forest extent and biomass over the same area. The components can be interpreted jointly as a noncompensating dashboard; if a single index is required, its valuation, normalization, weights, and permitted substitution must be explicit rather than hidden inside the unit. This approach is consistent with the separate physical ecosystem-extent and condition accounts in SEEA Ecosystem Accounting, with services and monetary presentations kept distinct (United Nations et al., 2021). Mapping such a scoreboard to the model's effective A_c and quality q still requires a declared observation operator and independent measurements; the NFA biocapacity total alone cannot supply it.

The empirical program that follows is a flux reconstruction, not a curve fit. The observation operator of Section 2.3 is a hierarchy: NFA land-type biocapacity with physical area and yield factors where obtainable; FAOSTAT area and yields as independent external proxies (used here); remote-sensing land cover and historical reconstructions for validation and extension (SI §S5.4). Forest area is a sentinel proxy for A_c, not the whole capital book; capital-land regeneration is unobserved in the accounts and would require growing-stock or net-annual-increment data. The proxies do not solve global-hectare identi-

fiability; they make the composition mechanism an independently anchored, falsifiable statement rather than an asserted one.

4.3. Policy operators

Three operators follow from the typed-floor structure and are developed in SI §S6. Freezing conversion – capping A_f or flooring A_c – is the single operator that guarantees $A_c \geq A_c^{\min}$; a governor that observes only the composite B cannot guarantee a capital-land floor, and the failure is informational rather than dynamical. Restoration is protective only when surplus-gated, and a deficit-gated flow can deepen the shortfall it is meant to relieve; this is the protective, as opposed to mobilizing, institutional signature of the delay companion (Abaee, 2026a). A Half-Earth reservation (Wilson, 2016), $E \leq \sigma B$ with $\sigma < 1$, succeeds without liquidation on the nominal path, but this is a nominal-path consequence until it is certified against a declared disturbance class under a preregistered retention rule, and we state it as such.

4.4. Relation to companion work

The accounting layer – the closed typed-flux ledger, conservation and donor-limitation rules, depletion arithmetic and the no-non-negative-weighting aggregation obstruction – is the object of the ledger companion (Abaee, 2026c), which owns the static layer, cedes delay dynamics to its companions and registers a two-pool dynamical model as an open gap; the present paper supplies that minimal dynamical realization for the global-hectare case. The aggregation theorem that places B at the weak extreme of the aggregator family is from the assessment companion (Abaee, 2026b), and the institutional-delay regime is from the governance companion (Abaee, 2026a); the delays here are ecological (regeneration, demography), not institutional. We cite, and do not re-derive, those results.

4.5. Limitations

The capital land is an effective class, not a single land type; the single regeneration timescale, the composition diagnostic and the ceiling are effective-parameter statements, and the full vector decomposition is cited rather than re-derived. Most parameters are representative rather than estimated; b_c, $b_{G,c}$ and ρ_c are not identifiable from a cross-section, and the regeneration anchors bound a band rather than calibrate ρ_c (the anchors concern the regeneration rate, not the lag τ_g). The per-capita footprint is held constant to isolate the

stock–flow–demand feedbacks; endogenizing e is an extension. The composition window is bounded and demand-driven; without a sustained shortfall or a technology/demand feedback it closes at the balance point. Stability at $S = 0$ is one-sided because the positive part is exact. Drawdown thresholds are tolerance-dependent bands; a small margin next to that band is reported as such and not as a sharp boundary. The debt-repayment rate is set physical and non-singular ($\eta = 0.05\,\mathrm{yr}^{-1}$); the $\eta \to 0$ case removes the debt equilibrium and is excluded by a floor, and outcomes are insensitive to η over a modest positive range. The accounts do not separate yield from area, which is the identifiability limitation the paper is about rather than an estimation error. Finally, the delayed-regeneration term is implemented at the current quality in the deposited solver (Section 2.4); we have verified that applying the delay changes no reported quantity by more than 1%, and the corrected variant is deposited alongside.

5. Conclusions

A single aggregate of annual bioproductive supply can conceal the composition of what is growing, and the concealment is structural. In a minimal two-land model with conserved area, a separate quality state and demand-driven conversion, the conversion contribution to aggregate change is positive when the fast-land yield exceeds the per-hectare value of capital land, while the active area transfer reduces capital-land area A_c; other terms determine whether the net aggregate rises or falls (Proposition 1). An aggregate-only observation does not locally identify the capital-land area loss, and independent fast-book proxies identify at most a measured fast book and an accounting residual. The drawdown of capital-land area A_c under sustained demand is a floor collision at a demand ceiling, not a catastrophic shift: the deficit-side dynamics are affine, the leading eigenvalue is constant along the branch, there is no fold, no hysteresis and no critical slowing down, and the drawdown depth is a continuous function of demand (Proposition 2). The mask is bounded because technology lifts the fast yield by at most a fixed factor whereas debt erosion is limited only by the sustained shortfall (Proposition 3), and recovery requires an explicit restoration flow whose gate sign matters. In the measured National Footprint and Biocapacity Accounts for 1961–2022, aggregate biocapacity grew 23% while per-capita biocapacity halved, the exact Divisia decomposition attributes the growth to cropland yield with a negative non-cropland residual, and the land-type accounts show

cropland biocapacity growing 2.85× against a flat non-cropland book that is itself the net of declining forest, grazing and fishing biocapacity and rising built-up land. These results establish an identifiability property of aggregate biocapacity and a measured composition pattern; they are not a forecast and do not show a measured decline in the model's effective capital-land area A_c, still less in any specific physical stock. The practical reporting rule has two parts: report NFA components alongside any aggregate, and track heterogeneous physical assets in a component-level scoreboard with explicit units, boundaries, condition measures, and separate flows. Do not collapse unlike stocks into an unweighted sum; if a single scalar is required, declare its valuation, weights, and permitted substitution. Neither the NFA aggregate nor the physical scoreboard identifies the model's effective A_c without an observation operator. A modern economy can run for decades on a harvest that looks healthy while the orchard behind it quietly shrinks; the harvest number will not say so, and only the two-book series can.

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Declarations

Data availability. The prior v43 code/data deposit is Zenodo record 22554480. The corrected v44 reproducibility package, including the solver, controlled-wave tests, generated metrics, figures, and manuscript/SI sources, is available at https://github.com/MIKEAA2020/general-sustainability/tree/main/agent%202%20productivity%20illusion/manuscript_ECOMOD_v44_src. This study reports no new empirical dataset; its empirical inputs are published values (NFA 2025 edition and FAOSTAT). The scripts that regenerate Fig. 1 and Table 2 accompany the Supplementary Information.

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